

Careflow Healthcare System

Comprehensive Engineering & Architecture Project Report

- Flutter Web & Mobile
- Firebase Authentication
- Cloud Firestore
- Material Design 3
- Multi-Role Architecture

1. Executive Summary

Careflow is an advanced, cross-platform healthcare management system engineered with **Flutter** and **Firebase**. The application delivers a unified, secure digital platform specifically tailored for three primary clinical stakeholders: **Patients**, **Doctors/Medical Staff**, and **Hospital Administrators**.

Careflow digitizes end-to-end clinical operations including online appointment scheduling, real-time consultation queues, patient health record management, medical staff directories, and administrative analytics. Built targeting modern Web and Mobile platforms, Careflow combines reactive state management with resilient offline handling.

2. Technology Stack & Environment

Domain	Technology / Library	Version / Specs	Operational Purpose
Frontend Framework	Flutter SDK	v3.13+	Cross-platform UI rendering for Web, iOS, macOS, Android
Language	Dart	v3.13+	Strongly-typed, asynchronous client execution
Authentication	firebase_auth	v6.1.3	Email/Password and Google OAuth Provider Authentication
Database Engine	cloud_firestore	v6.1.1	NoSQL real-time document store & cloud synchronization
Core SDK	firebase_core	v4.2.1	Platform initialization & Firebase Web SDK bridge
Design System	Material Design 3	Flutter Material	Responsive layout design system with curated palette

3. Multi-Role System Architecture

Careflow features a role-aware dynamic interface structure. Upon authentication, users are seamlessly routed to their specialized workspace based on permissions stored in Firestore.

Patient Workspace
Interactive appointment booking, active prescription tracking, care team lookup, and personal medical history timelines.

Doctor Workspace
Daily clinical appointment queue, visit approval workflows, patient medical chart access, and prescription writing tools.

Admin Control Center
Global patient directory, medical staff onboarding, security permission audits, and system performance metrics.

4. Key Functional Modules

4.1 Patient Management

- **Smart Booking Modal:** Select doctor, department specialty, preferred time slot, and specify visit symptoms.
- **My Appointments Queue:** View live appointment status (Confirmed, Pending, Completed) with one-click cancellation.
- **Care Team Catalog:** Browse available specialists by experience, department, and availability schedule.
- **Health Records Repository:** Access historical ECG reports, lab diagnostics, and active medications.

4.2 Doctor & Clinical Workflows

- **Schedule Approvals:** Review incoming patient booking requests and trigger Approve or Complete actions.
- **Patient Chart Access:** Review patient age, gender, blood group, chronic conditions, and prescription history.
- **Clinical Documentation:** Modal interface to append diagnosis notes and update patient treatment plans.

4.3 Administrative Operations

- **Medical Staff Onboarding:** Add new doctors to the hospital database with custom specialties and availability.
- **Patient Registration:** Manual patient profile registration for walk-in clinical admissions.
- **Real-time Security Stream:** System access logs, audit feeds, and platform performance monitoring.

5. Database Schema & Data Models

5.1 Collection: users

Field Name	Data Type	Description
uid	String (Document ID)	Unique identifier bound to Firebase Auth UID
email	String	User email address
displayName	String?	User's full name
role	String	System role: patient, doctor, or admin
createdAt	Timestamp	Server timestamp of registration

5.2 Collection: appointments

Field Name	Data Type	Description
id	String (Document ID)	Unique appointment identifier
patient	String	Patient full name
doctor	String	Attending physician name
specialty	String	Medical specialty / department
status	String	Lifecycle status: Pending, Confirmed, Completed, Cancelled
notes	String	Patient visit reason or clinical notes

6. Software Security & Resiliency

Careflow incorporates robust error handling and defensive programming patterns:

- **Resilient Offline Architecture:** All Cloud Firestore API calls are wrapped in try/catch blocks. In the event of network disruption or unprovisioned cloud databases, authentication completes smoothly via localized state fallback.
- **Specific Exception Handling:** Catches FirebaseAuthException to render specific user feedback (e.g. invalid password, account existing, network failure).
- **Role Authorization:** Strict control flow prevents unauthorized navigation to admin or clinical routes.

7. Verification & Build Results

```
$ flutter analyze Analyzing healthcare_system... No issues found! (ran in 1.5s) $ flutter
build web Compiling lib/main.dart for the Web... ✓ Built build/web
```

Verification Status: **ANALYSIS PASSED (0 ERRORS)** **WEB BUILD SUCCEEDED**
